6. Seaweed can contain high levels of iodine.

(a) One type of seaweed contains 0·133 g of iodine per kilogram of seaweed.

The World Health Organisation recommends a daily intake of iodine of 0·15 mg.

Calculate the mass of seaweed that would provide the recommended daily intake. **2**

(b) An experiment was carried out to determine the quantity of iodine in a sample of dried seaweed.

(i) The first step in this process involves burning an accurately known mass of dried seaweed to ash.

Describe the steps involved in measuring the mass of the seaweed by difference. **1**

(ii) The seaweed ash contains iodide ions. These react with hydrogen peroxide in acid conditions.

$$H_2O_2(aq) \quad + \quad 2I^-(aq) \quad + \quad 2H^+(aq) \quad \rightarrow \quad 2H_2O(\ell) \quad + \quad I_2(aq)$$

Identify the reducing agent in this reaction. **1**

6. (b) (continued)

(iii) The released iodine reacts with sodium thiosulfate, $Na_2S_2O_3$.

A standard solution of sodium thiosulfate, $Na_2S_2O_3$, is required to react with the released iodine.

State what is meant by a standard solution. **1**

(iv) For this sample of seaweed, 0·00026 moles of sodium thiosulfate were required to fully react with the released iodine, I_2.

$$I_2(aq) \quad + \quad 2Na_2S_2O_3(aq) \quad \rightarrow \quad 2NaI(aq) \quad + \quad Na_2S_4O_6(aq)$$

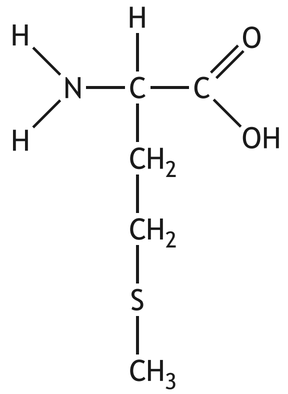
(A) Calculate the number of moles of iodine required to react with 0·00026 moles of sodium thiosulfate. **1**

(B) Using your answer to part (A), calculate the mass, in g, of iodine in the seaweed sample. **1**

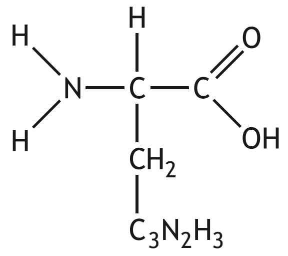
6. (continued)

(c) Seaweed is a major component of the diet of sheep living on the island of North Ronaldsay.

Sheep wool is made mainly of a protein. This protein contains the essential amino acids methionine and histidine.



methionine



histidine

(i) State what is meant by an essential amino acid. **1**

(ii) When two amino acids are joined together by a peptide link, a dipeptide is formed.

Draw a structural formula for the dipeptide formed from methionine and histidine. **1**